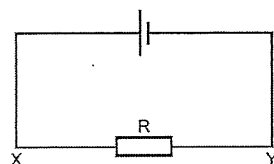


Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs

1. [IJC/H2/Prelims 2010/P1/25]

The current in the circuit is 4.8 A.

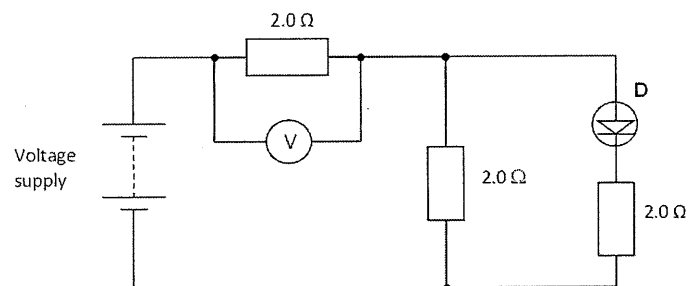


What is the rate of flow and the direction of flow of electrons through the resistor R?

	rate of flow / s^{-1}	direction of flow
A	3.0×10^{19}	X to Y
B	6.0×10^{18}	X to Y
C	3.0×10^{19}	Y to X
D	6.0×10^{18}	Y to X

2. [MJC/H2/Prelims 2010/P1/25]

In the circuit diagram below, D is an ideal diode. The voltage supply has negligible internal resistance and the voltmeter reads 12 V.



If the connections to the terminals of the voltage supply are reversed, the voltmeter reading would be

- A 6.0 V B 8.0 V C 9.0 V D 18 V

3. [NYJC/H2/Prelims 2010/P1/26]

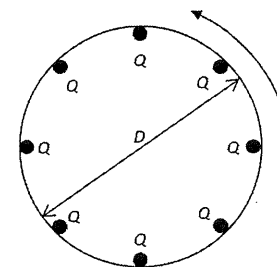
There are 1.3×10^{19} electrons passing through a point in a series circuit in 1.0 hour.

If the potential difference across a resistor connected in series to the source is 10 kV, what is the power dissipated in the resistor?

- A 3.6 W B 5.1 W C 5.8 W D 21 kW

4. [RI/H2/Prelims 2010/P1/25]

Eight small conductors of charge Q are placed on the edge of an insulating disc of diameter D . The angular frequency of rotation of the disc is ω .



What is the equivalent electric current at the edge of the disc?

- A $\frac{4Q\omega}{\pi}$ B $\frac{8Q\omega}{\pi D}$ C $8Q\omega$ D $\frac{16Q\pi}{\omega}$

5. [VJC/H2/Prelims 2019/P1/16]

A metal wire of cross sectional area 1.0 mm^2 has a conducting electron density of $6.0 \times 10^{28} \text{ m}^{-3}$.

If the wire is carrying 1.0 A of current, what is the approximate average drift speed of the conducting electrons along the wire?

- A 1.0 cm s^{-1} B 1.0 mm s^{-1} C 0.10 mm s^{-1} D 0.010 mm s^{-1}

6. [YJC/H2/Prelims 2010/P1/23]

A 10Ω resistor is connected across a cell of e.m.f. 6.0 V and internal resistance 2.0Ω .

How much heat energy is dissipated by the resistor when 2.0 C of charge flows through it?

- A 2.0 J B 10 J C 12 J D 40 J

7. [YJC/H2/Prelims 2019/P1/22]

The current I in a metal wire is given by the expression:

$$I = A n v q$$

What does the symbol n represent?

- A the number of atoms per unit volume of the metal
 B the number of free electrons per atom in the metal
 C the number of free electrons per unit volume of the metal
 D the total number of electrons per unit volume of the metal

8. [NYJC/H2/Prelims 2019/P1/20]

A bulb lights up almost immediately when the switch is closed although the drift velocity of the electrons in the circuit is low at the start. This occurs because

- A the resistance of the wires is very low.
- B the electrons take a very short time to reach the bulb.
- C all free electrons drift simultaneously in the circuit.
- D the number density of electrons is high.

9. [AJC/H2/Prelims 2010/P1/33]

A signal generator can either produce a sinusoidal wave as in Fig. 9.1 or a square wave as in Fig. 9.2 with the same peak value of e.m.f. as illustrated below.

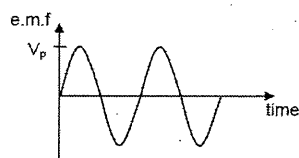


Fig. 9.1

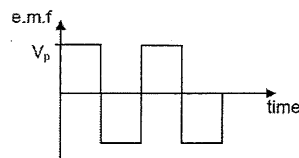


Fig. 9.2

When the signals are applied to a resistor R , what is the ratio of $\frac{\text{mean power of sinusoidal wave}}{\text{mean power of square wave}}$?

- A 2
- B $\frac{1}{2}$
- C 4
- D $\frac{1}{4}$

10. [CJC/H2/Prelims 2010/P1]

When an alternating current flows through a resistor of $5.0\ \Omega$, heat is dissipated at a rate of $20\ \text{W}$. What is the peak value of this alternating current?

- A 1.41 A
- B 2.00 A
- C 2.83 A
- D 4.00 A

11. [CJC/H2/Prelims 2010/P1/32]

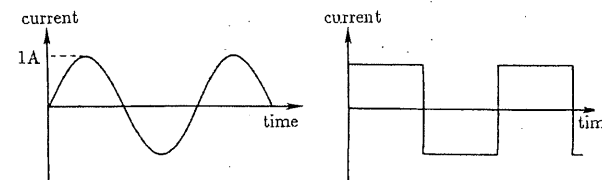
A sinusoidal alternating current has period T . The r.m.s. value of the current in a resistor is I and the mean power dissipated in the resistor is P .

Which statement is correct?

- A The frequency is $\frac{2\pi}{T}$.
- B The maximum power dissipated in the resistor is $P\sqrt{2}$.
- C The peak current is $\frac{I}{\sqrt{2}}$.
- D The r.m.s. voltage is $\frac{P}{I}$.

12. [Dunman High/H2/Prelims 2010/P1/32]

Alternating current of sinusoidal and square waveforms, as shown below, flow in turn through a resistor.



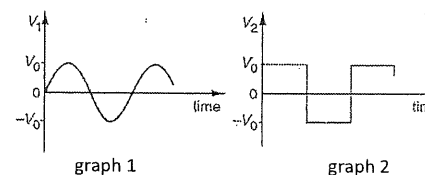
The power dissipated is the same in each case.

Given that the peak value of the sinusoidal current is $1.0\ \text{A}$, what is the peak value, in A, of the square waveform?

- A 2
- B $\sqrt{2}$
- C $\frac{1}{\sqrt{2}}$
- D $\frac{1}{2}$

13. [IJC/H2/Prelims 2010/P1/32]

A sinusoidal potential difference V_1 shown in graph 1 produces heat at a mean rate W when applied across a resistor R .



graph 1

graph 2

What is the mean rate of production of heat when the square-wave potential difference V_2 shown in graph 2 is applied across the resistor?

- A $\frac{W}{2}$
- B $\frac{W}{\sqrt{2}}$
- C $W\sqrt{2}$
- D $2W$

14. [MI/H2/Prelims 2010/P1/33]

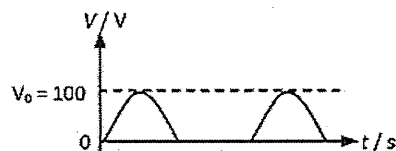
A sinusoidal alternating current flows through a resistor.

What is the mean power dissipated in the resistor?

- A $\frac{1}{\sqrt{2}}(\text{maximum current})^2$
- B $\frac{1}{\sqrt{2}}(\text{maximum power})$
- C $\frac{1}{2}(\text{maximum current})^2$
- D $\frac{1}{2}(\text{maximum power})$

15. [MI/H2/Prelims 2010/P1/34]

The half-wave rectification of an alternating current sinusoidal voltage of amplitude 100 V gives the waveform as shown.



What is the r.m.s. value of the rectified voltage?

- A 25 V B 50 V C 71 V D 100 V

16. [MJC/H2/Prelims 2010/P1/33]

A sinusoidal potential difference V_1 as shown in Fig. A, is applied across a resistor R and produces heat at a mean rate W .

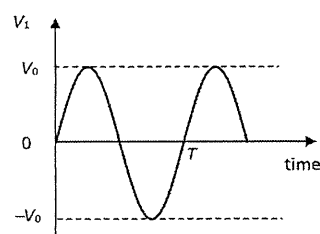


Fig. A

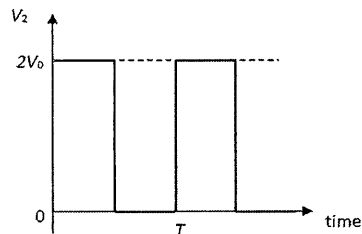


Fig. B

What is the mean rate of heat produced when another potential difference V_2 as shown in Fig. B is applied across the same resistor?

- A $\frac{1}{2}W$ B $2W$ C $4W$ D $8W$

17. [MJC/H2/Prelims 2010/P1/34]

An electric kettle has the following label:

Power : 2000 to 2400 W

Voltage : 220 to 240V

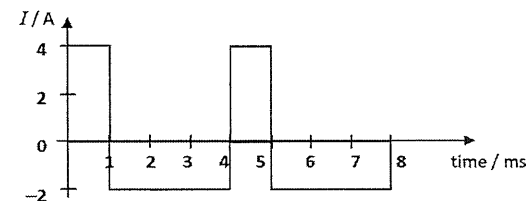
Frequency : 50 to 60 Hz

Which of the following is a probable expression of the current I that passes through the kettle when used in Singapore?

- A $I = 8.33\sin(315t)$ B $I = 10.9\sin(315t)$ C $I = 14.1\sin(375t)$ D $I = 16.0\sin(375t)$

18. [NJC/H2/Prelims 2010/P1/34]

The figure below shows the variation with time of a periodic current I .

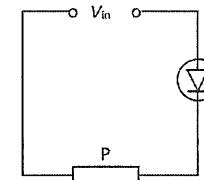
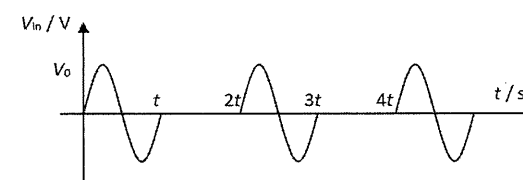


What is the root-mean-square value of the current?

- A 2.50 A B 2.65 A C 2.82 A D 3.00 A

19. [NYJC/H2/Prelims 2010/P1/33]

A rectifier is connected in series with load P and an alternating voltage supply as shown in the figure.



What is the r.m.s. voltage across P ?

- A $0.18 V_0$ B $0.35 V_0$ C $0.50 V_0$ D $0.71 V_0$

20. [NYJC/H2/Prelims 2010/P1/34]

An alternating current I / A varies with time t / s according to the equation

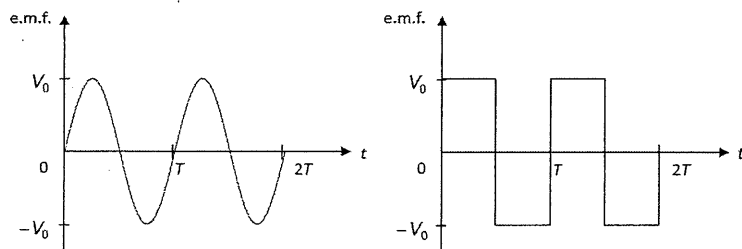
$$I / A = 5.00\sin(100\pi t)$$

What is the mean power developed by the current in a resistive load of resistance 10Ω ?

- A 50 W B 62.5 W C 125 W D 250 W

21. [PJC/H2/Prelims 2010/P1/33]

A signal generator produces either (a) a sinusoidal or (b) a square wave with the same peak value V_0 of e.m.f. as shown.



When the signals are applied to a resistor, what is the ratio of $\frac{\text{mean power of case (a)}}{\text{mean power of case (b)}}$?

- A $\frac{1}{4}$ B $\frac{1}{2}$ C $\frac{1}{\sqrt{2}}$ D 1

22. [PJC/H2/Prelims 2010/P1/34]

An alternating current of peak value 2 A and a steady direct current I flowing through identical resistors dissipate heat at equal rates.

What is the value of current I ?

- A $\frac{1}{2\sqrt{2}}$ A B $\sqrt{2}$ A C 2 A D $2\sqrt{2}$ A

23. [RI/H2/Prelims 2010/P1/33]

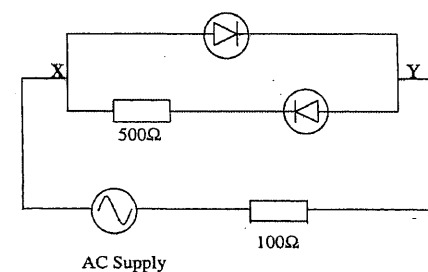
An alternating current I in amperes in a load resistor of 8.0Ω varies with time t in seconds according to the equation:

$$I/A = 5.0 \sin(\pi t)$$

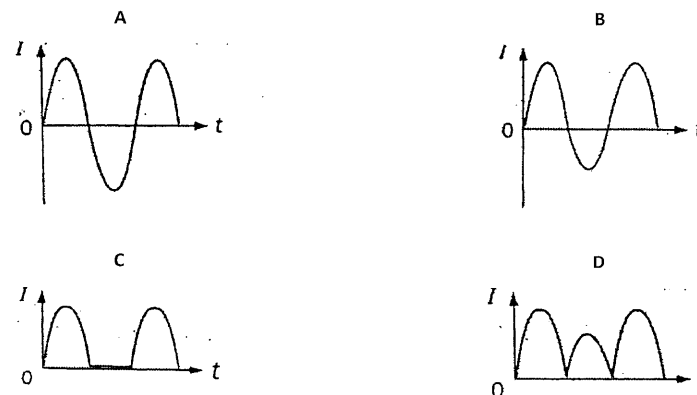
Which of the following is the mean power dissipated in the resistor?

- A 40 W B 50 W C 100 W D 200 W

24. [RVHS/H2/Prelims 2010/P1/32]



Which of the following graphs below represents the variation of current I with time t through XY of the circuit above?



25. [RVHS/H2/Prelims 2010/P1/33]

When an electric kettle is connected to an A.C. source, the power output is half of that when it is connected to a 10 V D.C. source.

What is the peak voltage of the A.C. source?

- A 5.0 V B 7.0 V C 10 V D 14 V

26. [SAJC/H2/Prelims 2010/P1/32]

A steady current I dissipates a certain power in a variable resistor. When a sinusoidal alternating current is used, the variable resistor has to be reduced to one quarter of its initial value to obtain the same power.

What is the peak value of the alternating current?

- A $\sqrt{2}I$ B $2I$ C $2\sqrt{2}I$ D $4\sqrt{2}I$

27. [TJC/H2/Prelims 2010/P1/34]

A sinusoidal current of peak value I_0 is passed through a resistor R with a diode connected in series with it. What is the average rate of heat dissipated in R ?

- A $0.25I_0^2R$ B $0.50I_0^2R$ C $0.71I_0^2R$ D $1.41I_0^2R$

28. [VJC/H2/Prelims 2010/P1/32]

The voltage V of an A.C. power supply, measured in volts, varies with time t , measured in seconds, as follows:

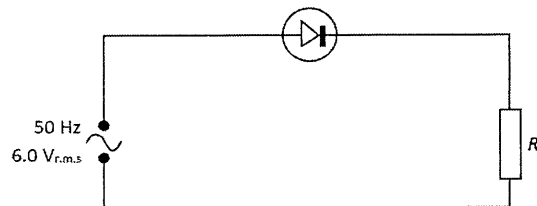
$$V = 3.0 \sin 380t.$$

What is the r.m.s. voltage and frequency of the power supply?

	r.m.s. voltage / V	frequency / Hz
A	2.1	60
B	2.1	380
C	3.0	60
D	3.0	380

29. YJC/H2/Prelims 2010/P1/30]

The figure below shows the circuit diagram for a half-wave rectifier. The supply to the rectifier is rated as 50 Hz, 6.0 V_{r.m.s.}.



What is the average power dissipated in the load resistor, $R = 2.0 \Omega$?

- A 9.0 W B 18 W C 25 W D 36 W

II. Structured Questions

1. [SAJC/H2/Prelims 2010/P2/5]*

- State an advantage of using alternating current for the transmission of electrical energy. [1]
- State why high voltage is used in the transmission of electrical energy. [1]
- An a.c. generator delivers power to a resistive load in a remote location over a cable transmission line of resistance 6.0 k Ω . The potential difference across the load is 80 kV, and the power delivered by the generator is 250 kW.
 - Show that the current flowing in the cables is 2.61 A. [2]
 - Calculate the total voltage drop across the transmission cables. [1]
 - Calculate the power dissipated in the transmission cables as thermal energy. [1]
 - Calculate the power delivered to the resistive load. [1]

[Total: 7]

2. [VJC/H2/Prelims 2010/P2/3]*

- A 240 V r.m.s. voltage is applied across the primary coil of a step-down transformer with a primary : secondary turns ratio of 20 : 1. Calculate the r.m.s. voltage induced across the secondary coils. [2]
- A 6.0 Ω resistor is placed across the secondary coil. Calculate the maximum power dissipated by the resistor. [3]
- It is desired that the voltage across the resistor be converted into a D.C. voltage. Complete the circuit in Fig. 4.1 to show how this could be achieved. Include the resistor in your circuit [1]

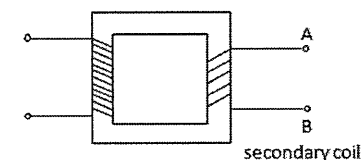


Fig. 4.1

- Sketch a graph of the voltage across the resistor in c. against time. [1]
- Suggest a disadvantage of the circuit in c.. [1]
- On Fig. 4.1, indicate the polarity of A and B when a current is flowing through the resistor [1]

[Total: 9]

3. [PJC/H2/Prelims 2010/P3/4]*

Fig. 5.1 shows a simple iron-cored transformer. The primary coil is connected to a 240 V r.m.s., 50 Hz power supply and the output voltage across the $6.0\ \Omega$ resistor in the secondary coil is 12 V r.m.s.

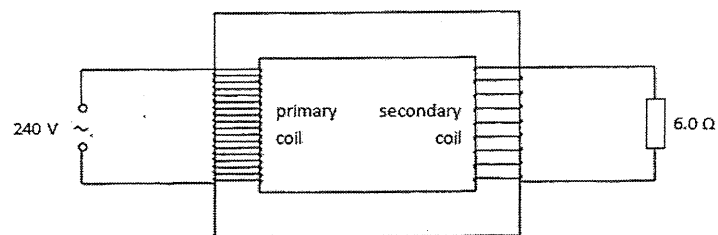


Fig. 5.1

- a.
 - i. State Faraday's law of electromagnetic induction. [1]
 - ii. Use Faraday's law to explain why a transformer will operate for an alternating input voltage but not for a direct voltage. [2]
- b.
 - i. Explain what is meant by *root-mean-square value* of an alternating voltage. [1]
 - ii. Calculate the peak output voltage. [1]
 - iii. Calculate the mean power dissipated in the $6.0\ \Omega$ resistor. [2]
 - iv. On the axes of Fig. 5.2, sketch a graph to show the variation with time t of the power P dissipated in the $6.0\ \Omega$ resistor over one period. [2]

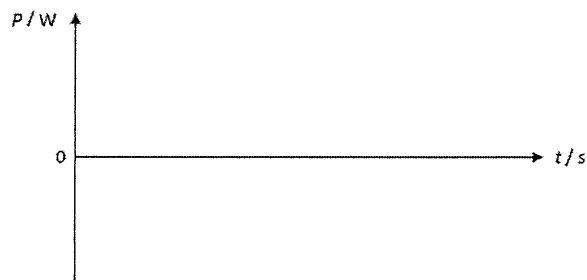
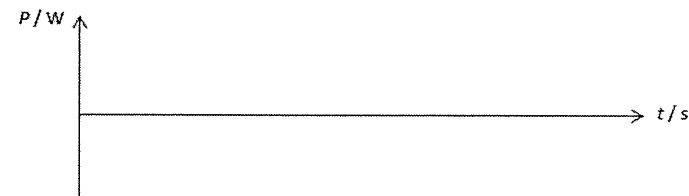


Fig. 5.2

[Total: 9]

4. [YJC/H2/Prelims 2010/P2/5]*

- a. A heating device is designed to operate on either an A.C. or D.C. power supply. The device has a resistance of $6.0\ \Omega$. Calculate the average power dissipated in the device when operating at
 - i. an A.C. supply of voltage 12.0 V, 50 Hz [2]
 - ii. a D.C. supply of voltage 12.0 V. [1]
- b. Draw the time t variation of the power P dissipated in the device for both the a.c. and d.c. supply on the same axes below. Mark values on both axes. [4]



- c. The alternating supply of voltage 12.0 V, 50 Hz is derived from the mains supply of voltage 230 V, 50 Hz using a transformer, assumed to have 100% efficiency. Calculate the primary r.m.s. current when the heating device is in use. [2]

[Total: 9]

III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	C	C	A	C	B	C	C	B	C	D	C	D	D	B

16	17	18	19	20	21	22	23	24	25	26	27	28	29
C	C	B	B	C	B	B	C	B	C	C	A	A	A

1 C Rate of flow of electrons $= \frac{4.8}{1.6 \times 10^{-19}} = 3.0 \times 10^{19} \text{ s}^{-1}$
Electrons flow from negative to positive terminal.

2 C Ideal diode is forward-biased $\rightarrow$ zero resistance $\rightarrow$ equivalent resistance of // arrangement is 1.0Ω
 $\rightarrow$ p.d. across // arrangement is 6.0 V
 $\rightarrow$ e.m.f. is thus 18 V

Diode is reverse-biased $\rightarrow$ infinite resistance $\rightarrow$ equivalent resistance of // arrangement is now 2.0Ω
Since resistor which voltmeter is connected across is also 2.0Ω , its p.d. $= 18 / 2 = 9.0 \text{ V}$

3 C $I = \frac{Ne}{t} = \frac{1.3 \times 10^{19} \times 1.6 \times 10^{-19}}{3600} = 5.778 \times 10^{-4} \text{ A}$
 $P = IV = 5.778 \times 10^{-4} (10000) = 5.8 \text{ W}$

4 A $\omega = \frac{2\pi}{T}$
 $I = \frac{8Q}{T} = \frac{8Q}{2\pi I \omega} = \frac{4Q\omega}{\pi}$

5 C $I = nevA$
 $1.0 = 6.0 \times 10^{28} (1.6 \times 10^{-19}) v \left(\frac{1.0}{1000^2} \right)$
 $v = 1.0 \times 10^{-4} \text{ m s}^{-1}$

6 B p.d. across 10Ω resistor $= \frac{10}{10+2} \times 6.0 = 5.0 \text{ V}$

Note: p.d. between two points in a circuit is $V = \frac{W}{Q}$

where W = electrical energy converted to other forms of energy (e.g. heat)

Q = net amount of charge that passes from one point to the other.

Heat dissipated by 10Ω resistor, $W = VQ = 5.0(2) = 10 \text{ J}$

7 C By definition, n is the number density of free charge carriers (in this case, electrons) in the metal wire.

8 C As explained by option C, it is the bulk movement of electrons that happens simultaneously, independent of drift velocity.

9 B Mean power $\langle P \rangle = \frac{V_{rms}^2}{R} \Rightarrow \langle P \rangle \propto V_{rms}^2$

mean power of sinusoidal wave $= \frac{\left(\frac{V_p}{\sqrt{2}} \right)^2}{(V_p)^2} = \frac{1}{2}$

10 C Mean power $\langle P \rangle = I_{rms}^2 R$
 $\left(\frac{I_o}{\sqrt{2}} \right)^2 \times 5 = 20$
 $I_o = 2.83 \text{ A}$

11 D Mean power dissipated $\langle P \rangle = I_{rms} V_{rms}$
 $V_{rms} = \frac{\langle P \rangle}{I_{rms}} = \frac{P}{I}$

12 C Mean power dissipated in R , $\langle P \rangle = I_{rms}^2 R$
 $\langle P \rangle = \left(\frac{1}{\sqrt{2}} \right)^2 R = I_o^2 R$
 $I_o = \frac{1}{\sqrt{2}} \text{ A}$

13 D Mean power $\langle P \rangle = \frac{V_{rms}^2}{R}$

For sinusoidal pd, $W = \frac{\left(\frac{V_o}{\sqrt{2}} \right)^2}{R} = \frac{V_o^2}{2R}$ For rectangular pd, $W' = \frac{V_o^2}{R} = 2W$

14 D Mean power dissipated $\langle P \rangle = I_{rms} V_{rms} = \left(\frac{I_o}{\sqrt{2}} \right) \left(\frac{V_o}{\sqrt{2}} \right) = \frac{I_o V_o}{2} = \frac{\text{max. power}}{2}$

15 B $V_{rms} = \sqrt{\left(\frac{100^2}{2} \right) \times \frac{T}{2}} = 50 \text{ V}$

16 C For sinusoidal pd, $W = \frac{\left(\frac{V_o}{\sqrt{2}} \right)^2}{R} = \frac{V_o^2}{2R}$

For rectangular pd, $W' = \frac{\left(\sqrt{\frac{4V_o^2 \times T/2}{T}} \right)^2}{R} = \frac{2V_o^2}{R}$

$W' = 4W$

17 C Angular frequency of power supply $\omega = 2\pi f$
Range of ω : 314 to 377 Hz

Mean power $\langle P \rangle = V_{rms} I_{rms} \Rightarrow I_{rms} = \frac{\langle P \rangle}{V_{rms}}$
Range of r.m.s. current: 8.33 to 10.91 A
Range of peak current: 11.78 A to 15.43 A

18 B The period $T = 4 \text{ ms}$

$I_{rms} = \sqrt{\frac{4^2 \times 1 + (-2)^2 \times 3}{4}} = 2.65 \text{ A}$

19 B Period of V_{in} , $T = 2t$

Across P, $V_{rms} = \sqrt{\frac{\left(\frac{V_o^2}{2} \right) \times \frac{t}{2}}{2t}} = 0.35V_o$

20 C Mean power $\langle P \rangle = I_{rms}^2 R = \left(\frac{5.00}{\sqrt{2}} \right)^2 \times 10 = 125 \text{ W}$

21 B Mean power $\langle P \rangle = \frac{V_{rms}^2}{R} \Rightarrow \langle P \rangle \propto V_{rms}^2$

$\frac{\text{mean power of (a)}}{\text{mean power of (b)}} = \frac{\left(\frac{V_o}{\sqrt{2}} \right)^2}{(V_o)^2} = \frac{1}{2}$

22 B Mean power $\langle P \rangle = \left(\frac{2.0}{\sqrt{2}} \right)^2 R = I^2 R$

$I = \frac{2.0}{\sqrt{2}} = \sqrt{2} \text{ A}$

23 C Mean power dissipated in resistor, $\langle P \rangle = \left(\frac{5.0}{\sqrt{2}} \right)^2 \times 8 = 100 \text{ W}$

24 B $I_{xy}(t) = \frac{V_{xy} \sin \omega t}{R}$

When V_{xy} is positive, top wire is forward biased but bottom wire is reverse biased. Current flows in top wire only from X to Y (positive).

When V_{xy} is negative, top wire is reverse biased but bottom wire is forward biased. Current flows in bottom wire only from Y to X (negative). But the magnitude of this current is less than the current when V_{xy} is positive due to the 500Ω resistor.

25 C For a.c. source, mean power $\langle P \rangle = \left(\frac{V_o}{\sqrt{2}} \right)^2 \frac{1}{R}$

$\left(\frac{V_o}{\sqrt{2}} \right)^2 \frac{1}{R} = \frac{1}{2} \left(\frac{10^2}{R} \right)$

$V_o = 10 \text{ V}$

26 C Mean power dissipated in R, $\langle P \rangle = I_{rms}^2 R$

For d.c. supply, $P = I^2 R$

For a.c. supply, $P = \left(\frac{I_o}{\sqrt{2}} \right)^2 \frac{R}{4}$

$\therefore \left(\frac{I_o}{\sqrt{2}} \right)^2 \frac{R}{4} = I^2 R \Rightarrow I_o = 2\sqrt{2}I$

27 A r.m.s. current through R, $I_{rms} = \sqrt{\frac{\left(\frac{I_o^2}{2} \right) \times \frac{T}{2}}{T}} = \frac{I_o}{2}$

average rate of heat dissipated in R $\langle P \rangle = \left(\frac{I_o}{2} \right)^2 R = 0.25I_o^2 R$

28 A $\omega = 2\pi f = 380$
 $\therefore f = 60.48 \text{ Hz}$

$V_{rms} = \frac{3.0}{\sqrt{2}} = 2.12 \text{ V}$

29 A

$$\text{r.m.s. voltage across } R, V_{rms} = \sqrt{\left(\frac{V_o^2}{2}\right) \times \frac{T}{2}} = \frac{V_o}{2}$$

$$\therefore \text{average power dissipated in } R, \langle P \rangle = \frac{\left(\frac{6.0 \times \sqrt{2}}{2}\right)^2}{2} = 9.0 \text{ W}$$

Structured Questions

1. [SAJC/H2/Prelims 2010/P2/5]

- a. A.C. can be easily stepped up or down to the desired voltage using a transformer.
b. To reduce power loss (as Joule heating) as the transmission current is reduced.

$$\begin{aligned} P_{\text{delivered}} &= P_{\text{output}} + P_{\text{lost}} \\ &= I_c V_{\text{load}} + I_c^2 R_c \\ \therefore 250 \times 10^3 &= I_c \times (80 \times 10^3) + I_c^2 \times (6 \times 10^3) \end{aligned}$$

$$\text{Solving, we obtain } I_c = 2.61 \text{ A}$$

$$\text{ii. Total voltage drop across cables } = I_c R_c = (2.61) \times (6 \times 10^3) = 15.7 \text{ kV}$$

$$\text{iii. Power dissipated in cables } = I_c^2 R_c = (2.61)^2 \times (6 \times 10^3) = 40.9 \text{ kW}$$

$$\text{iv. Power delivered to load } = I_c V_{\text{load}} = (2.61) \times (80 \times 10^3) = 209 \text{ kW}$$

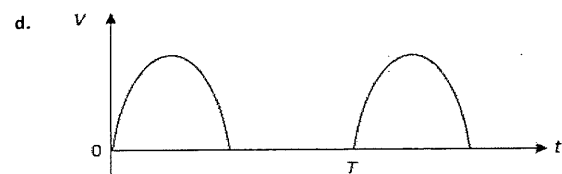
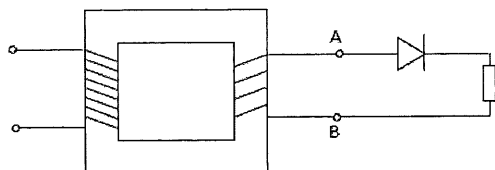
2. [VJC/H2/Prelims 2010/P2/3]

$$\text{a. } (V_{r.m.s.})_S = \frac{N_S}{N_P} (V_{r.m.s.})_P = \frac{1}{20} \times (240) = 12 \text{ V}$$

- b. Maximum power is dissipated when voltage is maximum.

$$P_{\text{max}} = \frac{(V_{\text{max}})^2}{R} = \frac{(\sqrt{2} \times V_{r.m.s.})^2}{R} = \frac{2 \times (12)^2}{6.0} = 48 \text{ W}$$

- c. Inclusion of a diode



- e. Half of the power supplied is not transferred to the resistor.

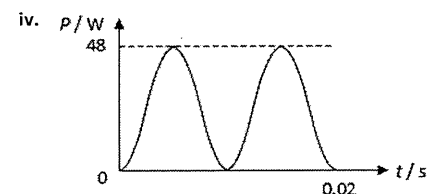
- f. A: (+); B: (–)

3. [PJC/H2/Prelims 2010/P3/4]

- a. i. Faraday's law states that the induced e.m.f. is proportional to the rate of change of magnetic flux linkage.
ii. An alternating input voltage will result in a constantly changing magnetic flux linkage in the secondary coil. An e.m.f. will be induced in the secondary coil.
A direct input voltage will result in a constant magnetic flux linkage in the secondary coil. Thus, no e.m.f. will be induced in the secondary coil.
- b. i. Root-mean-square value of an alternating voltage is the steady direct voltage that, when applied across a given resistance, will convert electrical energy to other forms of energy at the same rate as the alternating voltage.

$$\text{ii. } V_{\text{peak}} = V_{r.m.s.} \sqrt{2} = (12) \times \sqrt{2} = 16.97 = 17 \text{ V}$$

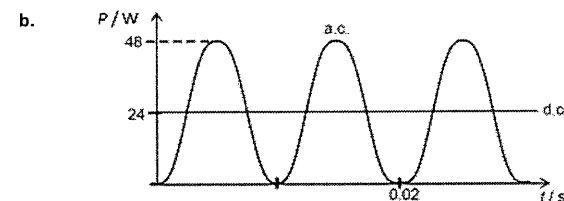
$$\text{iii. } P_{\text{ave}} = \frac{V_{r.m.s.}^2}{R} = \frac{(12)^2}{6.0} = 24 \text{ W}$$



4. [YJC/H2/Prelims 2010/P2/5]

$$\text{a. i. } P = \frac{V^2}{R} = \frac{(12.0)^2}{6.0} = 24 \text{ W}$$

$$\text{ii. } P = \frac{V^2}{R} = \frac{(12.0)^2}{6.0} = 24 \text{ W}$$



$$\text{c. } V_P I_P = V_S I_S$$

$$I_P = \frac{V_S I_S}{V_P} = \frac{(12.0) \left(\frac{12.0}{6.0} \right)}{230} = 0.10 \text{ A}$$

